Blessed G/Michael Catholic Secondary School

Semester 1 IT Final Examination for Grade12

Academic Year: 2024 / 2018 E.C. Time Allowed: 2:00'

Name: _____ **Grade & Sec:** Grade12 _____ **R.No:** _____

I. Write 'True' if the statement is correct and 'False' if it is incorrect.

____ **1.** A primary key in a relational database table can accept null values as long as other attributes are unique.

____ **2.** In symmetric-key encryption, both the sender and the receiver use the exact same secret key to encrypt and decrypt data.

____ **3.** The Network Layer (Layer 3) of the OSI model is responsible for logical addressing (IP addresses) and routing packets across subnets.

____ **4.** An IPv6 address consists of 32 bits divided into four decimal octets separated by dots.

____ **5.** In Object-Oriented Programming, Encapsulation refers to wrapping data (fields) and methods operating on that data into a single unit (class).

II. Choose the best answer among the given alternatives.

____ **6.** Which of the following SQL commands is categorized as Data Definition Language (DDL)?

A. SELECT **B.** INSERT **C.** CREATE **D.** UPDATE

____ **7.** In relational database design, a relation is in Second Normal Form (2NF) if it is in 1NF and:

A. Every non-prime attribute is fully functionally dependent on the entire primary key (no partial dependencies)

B. It contains no multi-valued attributes

C. There are no transitive functional dependencies

D. All foreign keys reference valid tables

____ **8.** Which SQL clause is used to filter records resulting from aggregate functions (such as COUNT, AVG, SUM)?

A. WHERE **B.** HAVING **C.** ORDER BY **D.** GROUP BY

____ **9.** In the TCP/IP suite, which protocol provides connection-oriented, reliable, and error-checked byte stream delivery?

A. UDP (User Datagram Protocol)

B. TCP (Transmission Control Protocol)

C. IP (Internet Protocol)

D. ICMP (Internet Control Message Protocol)

____ **10.** What standard port number is assigned by default to secure HyperText Transfer Protocol (HTTPS)?

A. 21 **B.** 53 **C.** 80 **D.** 443

____ **11.** Which social engineering attack tricks users into divulging sensitive credentials by masquerading as a trustworthy entity via fraudulent emails or fake websites?

A. Phishing **B.** Buffer Overflow

C. DDoS attack **D.** Cross-site scripting (XSS)

____ **12.** What type of malware encrypts a victim's files or system storage and demands payment to restore access?

A. Adware **B.** Ransomware **C.** Spyware **D.** Rootkit

____ **13.** Which data structure operates strictly on the Last-In, First-Out (LIFO) principle?

A. Queue **B.** Binary Search Tree

C. Stack **D.** Linked List

____ **14.** What is the worst-case time complexity of the Binary Search algorithm on a sorted array of n elements?

A. O(1) **B.** O(log n) **C.** O(n) **D.** $O(n^2)$

____ **15.** In HTML5, which tag is semantic for defining independent, self-contained content such as a blog post or news story?

A. **B.** <article> **C.** **D.** <content>

____ **16.** In JavaScript, which method is used to attach an event handler function to an HTML DOM element?

A. element.bindEvent()
B. element.addEventListener()
C. element.attachHandler()
D. element.listen()

____ **17.** Which OSI model layer is directly concerned with converting digital bitstreams into electrical, optical, or radio physical signals?

A. Physical Layer (Layer 1)
B. Data Link Layer (Layer 2)
C. Transport Layer (Layer 4)
D. Presentation Layer (Layer 6)

____ **18.** What database concept ensures that transactions are processed reliably according to ACID properties? What does ACID stand for?

A. Atomicity, Consistency, Isolation, Durability
B. Authentication, Control, Integrity, Data
C. Access, Concurrency, Indexing, Delivery
D. Allocation, Serialization, Iteration, Deletion

____ **19.** Which of the following IP addresses falls into a Private IPv4 range (RFC 1918) reserved for local area networks?

A. 8.8.8.8 **B.** 192.168.1.105 **C.** 208.67.222.222 **D.** 172.33.10.1

____ **20.** In software engineering, what phase of the SDLC (System Development Life Cycle) focuses on discovering user requirements and feasibility?

A. System Implementation
B. System Analysis & Requirements Engineering
C. Unit Testing
D. System Maintenance

____ **21.** In CSS, which layout module allows responsive one-dimensional arrangement of elements along either a row or column axis?

A. CSS Grid
B. Flexbox (Flexible Box Layout)
C. Float property
D. Absolute Positioning

____ **22.** Which network topology connects every computer directly to a central hub, switch, or router?

A. Bus topology **B.** Ring topology **C.** Star topology **D.** Mesh topology

____ **23.** What is the primary function of Domain Name System (DNS) in computer networks?

A. To assign dynamic IP addresses to connecting laptops
B. To translate human-readable domain names (e.g., example.com) into numerical IP addresses
C. To encrypt passwords stored in browser cookies
D. To detect malware packets entering network interfaces

____ **24.** In SQL, what statement is used to remove an entire database table along with its structure, constraints, and data permanently?

A. DELETE TABLE tablename;
B. TRUNCATE TABLE tablename;
C. DROP TABLE tablename;
D. REMOVE tablename;

____ **25.** Which principle of information security focuses on preventing unauthorized modification, tampering, or deletion of data?

A. Confidentiality
B. Integrity
C. Availability
D. Non-repudiation

____ **26.** What is the decimal equivalent of the binary number $(101101)_2$?

A. 41 **B.** 45 **C.** 53 **D.** 37

____ **27.** Which type of software license allows users to view, modify, and redistribute the source code freely?

A. Proprietary / Closed Source
B. Open Source Software (OSS)
C. Freeware without source
D. Commercial Shareware

____ **28.** In Object-Oriented Programming, the mechanism where a derived child class acquires properties and methods of an existing parent class is:

A. Polymorphism **B.** Inheritance **C.** Abstraction **D.** Overloading

____ **29.** What device operates at the Data Link Layer (Layer 2) and forwards frames based on physical MAC addresses?

A. Repeater **B.** Network Switch **C.** IP Router **D.** Passive Hub

____ **30.** In web applications, what client-side storage mechanism allows storing key-value pairs that persist across browser sessions until explicitly cleared?

A. sessionStorage **B.** localStorage **C.** PHP Session ID **D.** RAM cache